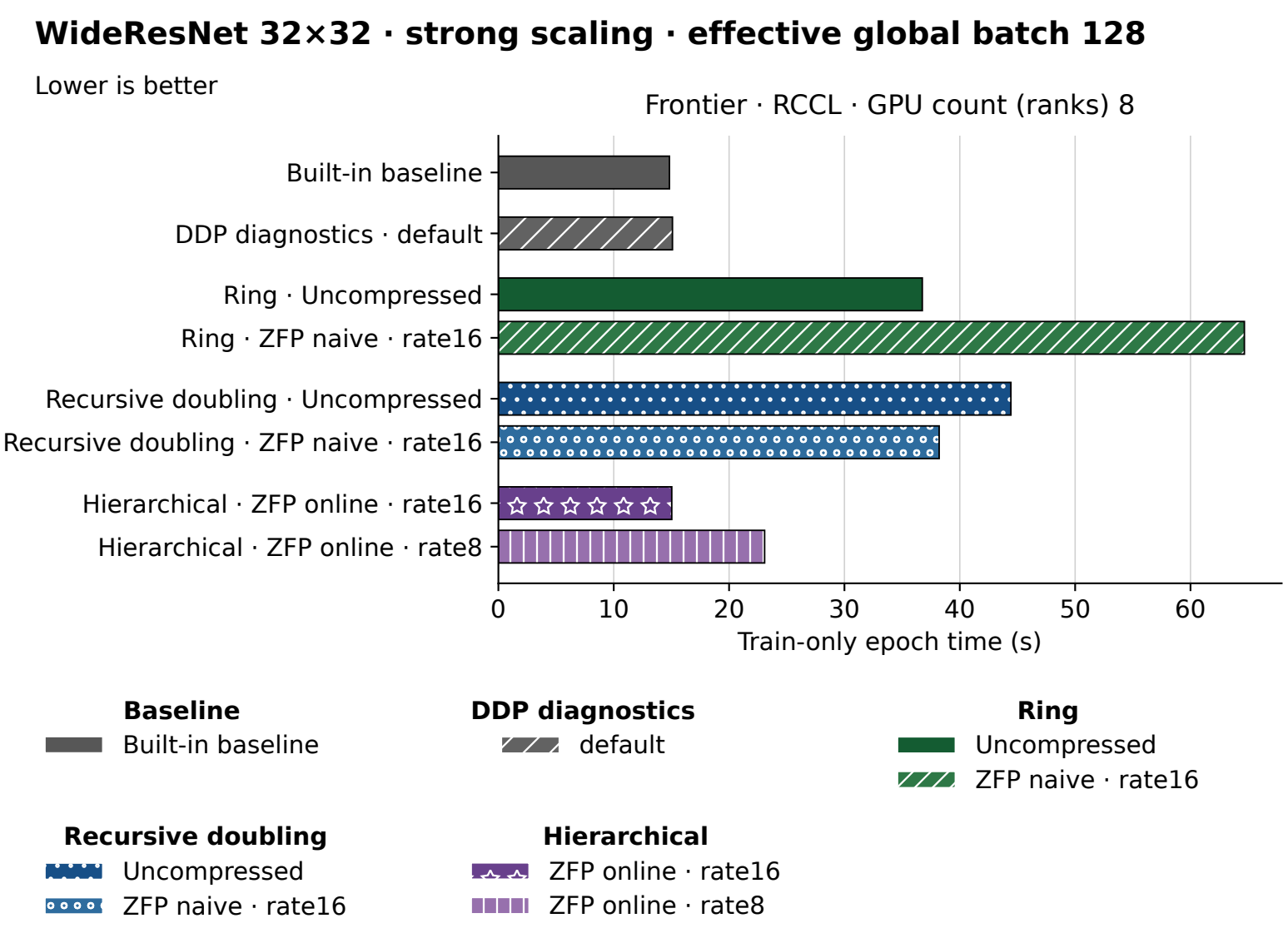

WideResNet 32×32 · strong scaling · effective global batch 128
Lower is better
Frontier · RCCL · GPU count (ranks) 8
Built-in baseline
DDP diagnostics · default
Ring · Uncompressed
Ring · ZFP naive · rate16
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Hierarchical · ZFP online · rate16
Hierarchical · ZFP online · rate8
0
10
20
30
40
50
60
Train-only epoch time (s)
Baseline
Built-in baseline
DDP diagnostics
default
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate